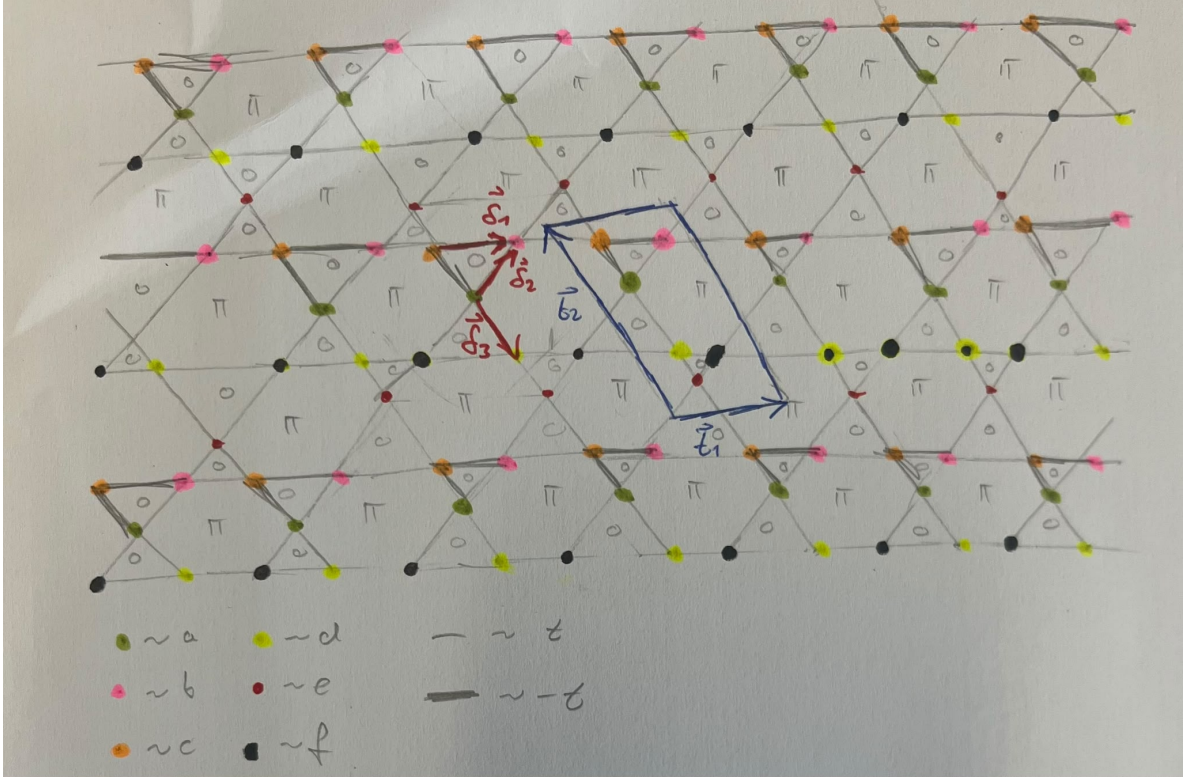


0.0.1 Real Space

This system is a monoclinic (Is this correct?) Bravais-lattice with a six site unit cell. It emerges from the bare Kagome lattice through unit cell doubling (the translational symmetry gets broken by the internal mean field).



The nearest neighbour vectors remain unchanged

$$\vec{\delta}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \vec{\delta}_2 = \frac{1}{2} \begin{pmatrix} 1 \\ \sqrt{3} \end{pmatrix} \quad \vec{\delta}_3 = \frac{1}{2} \begin{pmatrix} 1 \\ -\sqrt{3} \end{pmatrix}. \quad (1)$$

The new primitive translations are

$$\vec{t}_1 = \begin{pmatrix} 2 \\ 0 \end{pmatrix} \quad \vec{t}_2 = \begin{pmatrix} -2 \\ 2\sqrt{3} \end{pmatrix} \quad (2)$$

The real space (effective) tight-binding hamiltonian is

$$H = t \sum_{i\sigma} \left(-a_{i\sigma}^\dagger c_{i\sigma} - c_{i\sigma}^\dagger b_{i\sigma} + b_{i\sigma}^\dagger a_{i\sigma} + a_{i\sigma}^\dagger d_{i\sigma} + d_{i\sigma}^\dagger e_{i\sigma} + e_{i\sigma}^\dagger f_{i\sigma} + f_{i\sigma}^\dagger d_{i\sigma} \right. \\ \left. + b_{i\sigma}^\dagger c_{i+t_1;\sigma} + b_{i\sigma}^\dagger e_{i+t_1+t_2;\sigma} + f_{i\sigma}^\dagger d_{i+t_1;\sigma} + f_{i\sigma}^\dagger a_{i+t_1;\sigma} + e_{i\sigma}^\dagger c_{i-t_2;\sigma} + \text{h.c.} \right) \quad (3)$$

On the level of gauge-invariant quantities (/plaquette fluxes the system does not break any symmetry compared to the bare kagome lattice

0.0.2 Reciprocal Space

The reciprocal lattice vectors are

$$\vec{b}_1 = \pi \begin{pmatrix} 1 \\ 1/\sqrt{3} \end{pmatrix} \quad \vec{b}_2 = \pi \begin{pmatrix} 0 \\ 1/\sqrt{3} \end{pmatrix} \quad (4)$$

After a Fourier transform the k -space Hamiltonian reads

$$H = t \sum_{q,\sigma} \Psi_{q\sigma}^\dagger \begin{pmatrix} 0 & e^{iq_2} & -e^{-iq_3} & 0 & e^{-iq_2} & e^{iq_3} \\ e^{-iq_2} & 0 & 2i \sin(q_1) & e^{iq_2} & 0 & 0 \\ -e^{-iq_3} & -2i \sin(q_1) & 0 & e^{-iq_3} & 0 & 0 \\ 0 & e^{-iq_2} & e^{iq_3} & 0 & e^{iq_2} & e^{-iq_3} \\ e^{iq_2} & 0 & 0 & e^{-iq_2} & 0 & 2 \cos(q_1) \\ e^{-iq_3} & 0 & 0 & e^{iq_3} & 2 \cos(q_1) & 0 \end{pmatrix} \Psi_{q\sigma}, \quad (5)$$

where

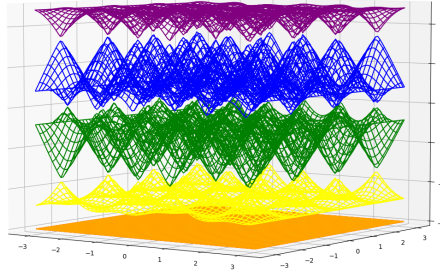
$$\Psi_{q\sigma}^\dagger = (a_{q\sigma}^\dagger \quad b_{q\sigma}^\dagger \quad c_{q\sigma}^\dagger \quad e_{q\sigma}^\dagger \quad f_{q\sigma}^\dagger \quad d_{q\sigma}^\dagger). \quad (6)$$

If I write this down somewhere serious, I should just label the sites differently.

I used the shorthand $q_i \equiv \vec{q} \cdot \vec{\delta}_i$.

0.0.3 dispersion and band trouching poiunts

The dispersion relation is shwon in the following picture

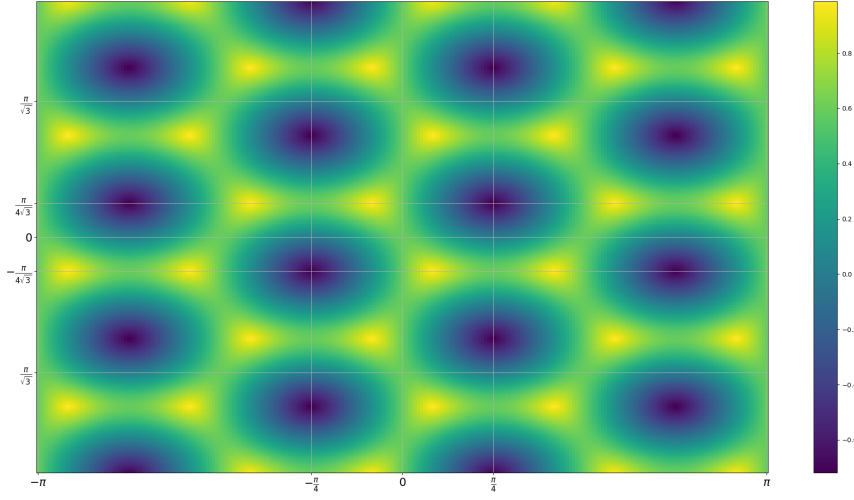


The lowest band (flat band) is doubly degenerate.

Directions on how to obtain the analytical expressions:

1. write down characteristic polynomial (probably the most annoying step)
2. Divide by the two constant roots corresponding to the flat bands (Polynomial of degree 4 remains)
3. Use symmetrie of remaining 4 bands (write as product of 2 quadratic polynomials; due to the symmetrie plane, one can guess p from the p-q-formula)
4. Solve remaining 2 quadratice equations
5. ?
6. Profit

At half filling the lowest three bands are filled, thus the Fermi-energy is between the yellow and green band.



Shown above is the 4th band (green band). Apparently there are two flavours of touching points with one of each in the first Brillouin zone. They are time reversal partners. From the numerical data these two touching points lie at

$$\vec{K} = \frac{\pi}{4} \begin{pmatrix} 1 \\ 1/\sqrt{3} \end{pmatrix}, \quad \vec{K}' = -\frac{\pi}{4} \begin{pmatrix} 1 \\ 1/\sqrt{3} \end{pmatrix}. \quad (7)$$

0.0.4 Linearization in reciprocal space and Projection

For now I only do the calculations for $\vec{K}$. Use

$$\vec{K} \cdot \vec{\delta}_1 = \frac{\pi}{4} \quad (8)$$

$$\vec{K} \cdot \vec{\delta}_2 = \frac{\pi}{4} \quad (9)$$

$$\vec{K} \cdot \vec{\delta}_3 = 0 \quad (10)$$

to obtain the linearized Hamiltonian in the vicinity of $\vec{K}$:

$$\begin{pmatrix} 0 & e^{i\frac{\pi}{4}}(1+ip_2) & -1+ip_3 & 0 & e^{-i\frac{\pi}{4}}(1-ip_2) & 1+ip_3 \\ e^{-i\frac{\pi}{4}}(1-ip_2) & 0 & \sqrt{2}i(1+p_1) & e^{i\frac{\pi}{4}}(1+ip_2) & 0 & 0 \\ -1-ip_3 & -\sqrt{2}i(1+p_1) & 0 & 1-ip_3 & 0 & 0 \\ 0 & e^{-i\frac{\pi}{4}}(1-ip_2) & 1+ip_3 & 0 & e^{i\frac{\pi}{4}}(1+ip_2) & 1-ip_3 \\ e^{i\frac{\pi}{4}}(1+ip_2) & 0 & 0 & e^{-i\frac{\pi}{4}}(1-ip_2) & 0 & \sqrt{2}(1-p_1) \\ 1-ip_3 & 0 & 0 & 1+ip_3 & \sqrt{2}(1-p_1) & 0 \end{pmatrix} \quad (11)$$

To keep the size reasonable maybe its a good idea to split this up into zeroth and first order in p. Might also be a good choice for storytelling purposes.

The shorthand means

$$p_i \equiv \vec{\delta}_i \cdot \vec{p},$$

also

$$e^{i\frac{\pi}{4}} = \frac{1+i}{\sqrt{2}}. \quad (12)$$

The eigenvalues for the zeroth order (/directly at the $\vec{K}$ -point) are

$$\begin{aligned} \varepsilon_1(\vec{K}) &= -2t & \varepsilon_2(\vec{K}) &= -2t & \varepsilon_3(\vec{K}) &= (1 - \sqrt{3})t \\ \varepsilon_4(\vec{K}) &= (1 - \sqrt{3})t & \varepsilon_5(\vec{K}) &= (1 + \sqrt{3})t & \varepsilon_6(\vec{K}) &= (1 + \sqrt{3})t \end{aligned} \quad (13)$$

ε_3 and ε_4 are the relevant ones.

I let mathematica calculate orthonormal eigenvectors for H at $\vec{K}$. The relevant ones (that is bands 3 and 4):

$$\vec{v}_3(\vec{K}) = \frac{1}{2\sqrt{6-3\sqrt{3}}} \left(\frac{1-\sqrt{3}}{2} + \frac{3-\sqrt{3}}{2}i, \quad \sqrt{2}(1-\sqrt{3}), \quad (-2+\sqrt{3})i, \quad \frac{1-\sqrt{3}}{2} + \frac{\sqrt{3}-3}{2}i, \quad 0, \quad 1 \right)^T \quad (14)$$

$$\vec{v}_4(\vec{K}) = \frac{1}{2\sqrt{6(7-4\sqrt{3})}} \left((3-2\sqrt{3}) + (-2+\sqrt{3})i, \quad 0, \quad i(1-\sqrt{3}), \quad 3-2\sqrt{3} + (2-\sqrt{3})i, \quad 2\sqrt{2}(2-\sqrt{3}), \quad 5-3\sqrt{3} \right)^T \quad (15)$$

The projections onto those vectors are

$$\begin{aligned} \vec{v}_3^H \cdot H_{\vec{K}} \cdot \vec{v}_3 &= (1 - \sqrt{3})t - \frac{2t}{\sqrt{3}}p_2 + \frac{(\sqrt{3}-1)t}{\sqrt{3}}p_3 \\ \vec{v}_4^H \cdot H_{\vec{K}} \cdot \vec{v}_4 &= (1 - \sqrt{3})t + \frac{2t}{\sqrt{3}}p_2 - \frac{(\sqrt{3}-1)t}{\sqrt{3}}p_3 \\ \vec{v}_3^H \cdot H_{\vec{K}} \cdot \vec{v}_4 &= -\frac{2t}{\sqrt{3}}p_2 - \frac{(3+\sqrt{3})t}{3}p_3 \\ \vec{v}_4^H \cdot H_{\vec{K}} \cdot \vec{v}_3 &= -\frac{2t}{\sqrt{3}}p_2 - \frac{(3+\sqrt{3})t}{3}p_3 \end{aligned} \quad (16)$$

I transform these to p_x and p_y :

$$\begin{aligned} \vec{v}_3^H \cdot H_{\vec{K}} \cdot \vec{v}_3 &= (1 - \sqrt{3})t + \frac{1-\sqrt{3}}{2}tp_x - \frac{1+\sqrt{3}}{2}tp_y \\ \vec{v}_4^H \cdot H_{\vec{K}} \cdot \vec{v}_4 &= (1 - \sqrt{3})t - \frac{1-\sqrt{3}}{2}tp_x + \frac{1+\sqrt{3}}{2}tp_y \\ \vec{v}_3^H \cdot H_{\vec{K}} \cdot \vec{v}_4 &= -\frac{1+\sqrt{3}}{2}tp_x - \frac{1-\sqrt{3}}{2}tp_y \\ \vec{v}_4^H \cdot H_{\vec{K}} \cdot \vec{v}_3 &= -\frac{1+\sqrt{3}}{2}tp_x - \frac{1-\sqrt{3}}{2}tp_y \end{aligned} \quad (17)$$

The dispersion of this projection is

$$\begin{aligned} \varepsilon_{3\vec{K}} &= (1 - \sqrt{3})t - \sqrt{2}t\sqrt{p_x^2 + p_y^2} \\ \varepsilon_{4\vec{K}} &= (1 - \sqrt{3})t + \sqrt{2}t\sqrt{p_x^2 + p_y^2} \end{aligned} \quad (18)$$

Most importantly

$$v_F = \sqrt{2}t \quad (19)$$

Judging from the pictures of the dispersion I didnt expect it to be so isotropical. Check this.

I follow the usual method defining

$$\begin{aligned} \vec{w}_3 &= \alpha \vec{v}_3 + \beta \vec{v}_4 \\ \vec{w}_4 &= \eta \vec{v}_3 + \gamma \vec{v}_4 \end{aligned} \quad (20)$$

Orthonormality is garuanted by

$$|\alpha|^2 + |\beta|^2 = 1 \quad (21)$$

$$|\eta|^2 + |\gamma|^2 = 1 \quad (22)$$

$$\alpha^* \eta + \beta^* \gamma = 0 \quad (23)$$

If my counting is correct, there are 8 parameters to begin with (real and imaginary parts), the equations above eliminate 1 and 1 and 2 (the last equation are really 2 equations, one for real and one for imaginary part), leaving behind 4 free parameters. I make the following Ansatz for the leftover parameters:

$$\begin{aligned} \alpha &= \cos(\phi/2) e^{i(\chi+\theta/2)} e^{i\kappa} \\ \beta &= \sin(\phi/2) e^{i(\chi+\theta/2)} \\ \eta &= \sin(\phi/2) e^{i(\chi-\theta/2)} e^{i\kappa} \\ \gamma &= -\cos(\phi/2) e^{i(\chi-\theta/2)} \end{aligned} \quad (24)$$

If you want to be fancy about it you can think of what kind of matrix describes the relation between the two bases and then find a suitable parametrization for it. If one thinks of these matrices in terms of groups, maybe there is a suffisticated argument to be made why $U(1)$ appears.. ϕ is a rotation angle, χ a global phase, θ the phase difference between the two states and κ state-internal phase. It is clear that all physical quantities can not depend on χ . At this point I might as well choose $\chi = 0$.

The following relations are impotant to derive the matrix elements in the rotated basis:

$$\begin{aligned} (2|\alpha|^2 - 1) &= \cos(\phi) \\ 2\text{Re}(\alpha^* \beta) &= \sin(\phi) \cos(\kappa) \\ \alpha^* \eta - \beta^* \gamma &= \sin(\phi) \cos(\theta) - i \sin(\phi) \sin(\theta) \\ \alpha^* \gamma + \beta^* \eta &= \sin(\kappa) \sin(\theta) - \cos(\phi) \cos(\kappa) \cos(\theta) \\ &\quad + i (\sin(\kappa) \cos(\theta) + \cos(\phi) \cos(\kappa) \sin(\theta)) \end{aligned}$$

Using these, in general (while neglecting the σ_0 -term),

$$\vec{w}_1^H \cdot H \cdot \vec{w}_1 \sim (\vec{v}_1^H \cdot H \cdot \vec{v}_1) \cos(\phi) + (\vec{v}_1^H \cdot H \cdot \vec{v}_2) \sin(\phi) \cos(\kappa) \quad (25)$$

$$\begin{aligned} \vec{w}_1^H \cdot H \cdot \vec{w}_2 &= (\vec{v}_1^H \cdot H \cdot \vec{v}_1) \sin(\phi) \cos(\theta) + (\vec{v}_1^H \cdot H \cdot \vec{v}_2) (\sin(\kappa) \sin(\theta) - \cos(\phi) \cos(\kappa) \cos(\theta)) \\ &\quad + i [- (\vec{v}_1^H \cdot H \cdot \vec{v}_1) \sin(\phi) \sin(\theta) + (\vec{v}_1^H \cdot H \cdot \vec{v}_2) (\sin(\kappa) \cos(\theta) + \cos(\phi) \cos(\kappa) \sin(\theta))] \end{aligned} \quad (26)$$

I find for the projection in the transformed basis (I again drop the σ_0 part as it transforms trivially):

$$\begin{aligned}
\frac{\vec{w}_3^H \cdot H_{\vec{K}} \cdot \vec{w}_3}{\sqrt{2}t} &\sim \frac{1}{\sqrt{8}} \left((1 - \sqrt{3}) \cos(\phi) - (1 + \sqrt{3}) \sin(\phi) \cos(\kappa) \right) p_x \\
&\quad + \frac{1}{\sqrt{8}} \left(-(1 + \sqrt{3}) \cos(\phi) - (1 - \sqrt{3}) \sin(\phi) \cos(\kappa) \right) p_y \\
\frac{\vec{w}_4^H \cdot H_{\vec{K}} \cdot \vec{w}_4}{\sqrt{2}t} &\sim \frac{1}{\sqrt{8}} \left(-(1 - \sqrt{3}) \cos(\phi) + (1 + \sqrt{3}) \sin(\phi) \cos(\kappa) \right) p_x \\
&\quad + \frac{1}{\sqrt{8}} \left((1 + \sqrt{3}) \cos(\phi) + (1 - \sqrt{3}) \sin(\phi) \cos(\kappa) \right) p_y \\
\frac{\vec{w}_3^H \cdot H_{\vec{K}} \cdot \vec{w}_4}{\sqrt{2}t} &= \frac{1}{\sqrt{8}} \left((1 - \sqrt{3}) \sin(\phi) \cos(\theta) - (1 + \sqrt{3}) [\sin(\kappa) \sin(\theta) - \cos(\phi) \cos(\kappa) \cos(\theta)] \right) p_x \\
&\quad + \frac{1}{\sqrt{8}} \left(-(1 + \sqrt{3}) \sin(\phi) \cos(\theta) - (1 - \sqrt{3}) [\sin(\kappa) \sin(\theta) - \cos(\phi) \cos(\kappa) \cos(\theta)] \right) p_y \\
&\quad - \frac{i}{\sqrt{8}} \left((1 - \sqrt{3}) \sin(\phi) \sin(\theta) + (1 + \sqrt{3}) [\sin(\kappa) \cos(\theta) + \cos(\phi) \cos(\kappa) \sin(\theta)] \right) p_x \\
&\quad - \frac{i}{\sqrt{8}} \left(-(1 + \sqrt{3}) \sin(\phi) \sin(\theta) + (1 - \sqrt{3}) [\sin(\kappa) \cos(\theta) + \cos(\phi) \cos(\kappa) \sin(\theta)] \right) p_y \\
\frac{\vec{w}_4^H \cdot H_{\vec{K}} \cdot \vec{w}_3}{\sqrt{2}t} &= \frac{1}{\sqrt{8}} \left((1 - \sqrt{3}) \sin(\phi) \cos(\theta) - (1 + \sqrt{3}) [\sin(\kappa) \sin(\theta) - \cos(\phi) \cos(\kappa) \cos(\theta)] \right) p_x \\
&\quad + \frac{1}{\sqrt{8}} \left(-(1 + \sqrt{3}) \sin(\phi) \cos(\theta) - (1 - \sqrt{3}) [\sin(\kappa) \sin(\theta) - \cos(\phi) \cos(\kappa) \cos(\theta)] \right) p_y \\
&\quad + \frac{i}{\sqrt{8}} \left((1 - \sqrt{3}) \sin(\phi) \sin(\theta) + (1 + \sqrt{3}) [\sin(\kappa) \cos(\theta) + \cos(\phi) \cos(\kappa) \sin(\theta)] \right) p_x \\
&\quad + \frac{i}{\sqrt{8}} \left(-(1 + \sqrt{3}) \sin(\phi) \sin(\theta) + (1 - \sqrt{3}) [\sin(\kappa) \cos(\theta) + \cos(\phi) \cos(\kappa) \sin(\theta)] \right) p_y
\end{aligned}$$

From these equations the $\sigma_{0,x,y,z}$ -components can be read of. I decided to solve for an equation of the form $H = \varepsilon_0 \sigma_0 + v_F(p_x \sigma_x + p_y \sigma_z)$. The corresponding angles are

$$\begin{aligned}
\cos(\phi) &= -\frac{1 + \sqrt{3}}{\sqrt{8}} \Rightarrow \phi = \frac{11}{12}\pi \\
\cos(\kappa) &= -\frac{1 - \sqrt{3}}{\sqrt{4 - 2\sqrt{3}}} = 1 \Rightarrow \kappa = 0 \\
\cos(\theta) &= \frac{1 - \sqrt{3}}{\sqrt{4 - 2\sqrt{3}}} = -1 \Rightarrow \theta = \pi.
\end{aligned} \tag{27}$$

$\sqrt{4 - 2\sqrt{3}} = \sqrt{3} - 1$:D. First I had lhs= $1 - \sqrt{3}$, and it took me around 3 hours to find the mistake.

Another solution corresponding to the form $H = \varepsilon_0 \sigma_0 + v_F(p_x \sigma_x + p_y \sigma_y)$ is

$$\begin{aligned}
\phi &= -\frac{\pi}{2} \\
\kappa &= -\frac{\pi}{2} \\
\theta &= \frac{5}{12}\pi
\end{aligned} \tag{28}$$

I will continue to work with this solution from now on. Typically I will employ the first solution as a sanity check.

0.0.5 Introduction of strain

The relevant equations are

$$t(\vec{\delta}_{ij})e^{i\vec{K}\cdot\vec{\delta}_{ij}} \approx te^{i\vec{K}\cdot\vec{\delta}_{ij}} \left(1 + i\vec{p}\cdot\vec{\delta}_{ij} - \beta \left(\vec{\delta}_{ij}\cdot\vec{u}\cdot\vec{\delta}_{ij} \right) + i \left(\vec{K}\cdot\vec{u}\cdot\vec{\delta}_{ij} \right) \right) \quad (29)$$

and

$$\begin{aligned} t(\vec{\delta}_{ij})e^{i\vec{q}\cdot\vec{\delta}_{ij}} + t(-\vec{\delta}_{ij})e^{-i\vec{K}\cdot\vec{\delta}_{ij}} &\approx 2t \cos(\vec{q}\cdot\vec{\delta}_{ij}) - 2t \sin(\vec{q}\cdot\vec{\delta}_{ij}) (\vec{p}\cdot\vec{\delta}_{ij}) \\ &\quad - 2\beta t \cos(\vec{K}\cdot\vec{\delta}_{ij}) (\vec{\delta}_{ij}\cdot\vec{u}\cdot\vec{\delta}_{ij}) \\ &\quad - 2t \sin(\vec{K}\cdot\vec{\delta}_{ij}) (\vec{K}\cdot\vec{u}\cdot\vec{\delta}_{ij}) \end{aligned} \quad (30)$$

and

$$\begin{aligned} t(\vec{\delta}_{ij})e^{i\vec{q}\cdot\vec{\delta}_{ij}} - t(-\vec{\delta}_{ij})e^{-i\vec{q}\cdot\vec{\delta}_{ij}} &\approx 2it \sin(\vec{K}\cdot\vec{\delta}_{ij}) + 2it \cos(\vec{K}\cdot\vec{\delta}_{ij}) (\vec{p}\cdot\vec{\delta}_{ij}) \\ &\quad - 2i\beta t \sin(\vec{K}\cdot\vec{\delta}_{ij}) (\vec{\delta}_{ij}\cdot\vec{u}\cdot\vec{\delta}_{ij}) \\ &\quad + 2it \cos(\vec{K}\cdot\vec{\delta}_{ij}) (\vec{K}\cdot\vec{u}\cdot\vec{\delta}_{ij}) \end{aligned} \quad (31)$$

I split the everything into two parts, the first one being proportional to β :

$$\langle \delta H_1 \rangle_{\vec{w}_1, \vec{w}_2} = -\frac{\beta}{2} (1 - \sqrt{3}) (u_{xx} + u_{yy}) \sigma_0. \quad (32)$$

There is also a β^0 -term that I shouldwrite down. However, this term can be gauged away so it not relevant for the physical discussion.

This means that to this order there is no effective vector potential but only an effective scalar potential.